

Project Report

Title

A Hybrid Deep-Learning Approach to Automated Misinformation Detection

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Course

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Introduction

Misinformation and fake news spread rapidly through social media and online platforms. Many users cannot easily verify whether a news statement is trustworthy or misleading. This project presents an NLP-based misinformation credibility detection system that predicts the truthfulness level of a statement using machine learning and deep learning techniques.

Instead of using only binary classification, the system uses six LIAR dataset labels: true, mostly-true, half-true, barely-true, false, and pants-fire. The project combines semantic

embeddings, transformer-based classification, soft risk scoring, and linguistic analysis to generate explainable credibility predictions.

The system was implemented using **Sentence-BERT, DistilBERT, Logistic Regression, PyTorch, and Streamlit**. The main focus of this report is the implementation, experimentation, model evaluation, and final system results.

The internet has transformed the way information is shared and consumed. Social media platforms, blogs, and online news websites allow information to spread rapidly across the world within seconds. Although this has improved communication and accessibility, it has also increased the spread of fake news and misinformation.

Misinformation can influence public opinion, create panic, damage reputations, and spread political or social confusion. In many situations, users cannot easily determine whether a statement is trustworthy. Manual fact-checking is time-consuming and requires human experts. Therefore, automated systems are needed to assist in identifying suspicious content.

This project focuses on building an automated misinformation credibility detection system using Natural Language Processing techniques. Instead of simple binary classification, the system predicts credibility using six truthfulness categories from the LIAR dataset. The system also provides credibility scores, risk scores, and linguistic explanations to improve interpretability.

The proposed **solution combines semantic machine learning methods with transformer-based deep learning approaches** to improve contextual understanding and prediction quality.

Project Objectives

The main objectives of this project are:

- To build an NLP-based misinformation detection system
- To classify statements using six LIAR truthfulness labels
- To compare machine learning and deep learning approaches
- To generate credibility and risk scores
- To analyze suspicious linguistic patterns
- To develop a user-friendly Streamlit interface

The main objectives of this project are:

- To develop an NLP-based misinformation detection system
- To classify statements using six LIAR truthfulness labels
- To implement semantic machine learning and transformer-based deep learning models
- To generate credibility scores and risk scores
- To analyze suspicious linguistic patterns
- To build a user-friendly Streamlit interface
- To compare traditional ML and deep learning approaches

Proposed System

The proposed system analyzes a news headline or statement entered by the user. The input text is preprocessed and passed through trained models to predict a truthfulness label.

The prediction probabilities are then used to calculate:

- Credibility Score
- Risk Score
- Final Credibility Category

In addition, the system performs linguistic analysis to detect suspicious writing styles such as excessive capitalization, emotional words, and clickbait phrases.

The final output is displayed through an interactive Streamlit interface.

Technical Architecture

The technical architecture of the system consists of the following major components:

1. User Interface Layer
2. Data Preprocessing Layer
3. Semantic Machine Learning Layer
4. Deep Learning Layer
5. Risk Scoring Module
6. Linguistic Analysis Module
7. Output Visualization Layer

System Flow:

User Input → Text Preprocessing → Model Prediction → Probability Generation → Risk Calculation → Linguistic Analysis → Final Credibility Output

Tools and Technologies Used

Layer	Technology
Programming Language	Python
Data Handling	Pandas, NumPy
NLP Preprocessing	NLTK
Semantic Embeddings	Sentence-BERT
Machine Learning	Scikit-learn
ML Classifier	Logistic Regression
Deep Learning	PyTorch
Transformer Model	DistilBERT
UI Framework	Streamlit
Training Platform	Google Colab GPU
Model Saving	Pickle, Joblib

Dataset Description

The project uses the LIAR dataset, which is a publicly available dataset containing short political statements labeled with truthfulness categories.

Dataset Files:

- train.tsv
- valid.tsv
- test.tsv

The dataset includes six labels:

Label

true
mostly-true

Label

half-true

barely-true

false

pants-fire

The dataset was cleaned and preprocessed before training.

Processed files:

- train_clean.csv
- valid_clean.csv
- test_clean.csv

Data Preprocessing

Data preprocessing was performed to improve text quality and prepare the dataset for model training.

The following preprocessing techniques were applied:

- Lowercasing text
- Removing unnecessary symbols
- Tokenization
- Stopword removal
- Basic text cleaning

NLTK was used for most preprocessing operations.

Machine Learning Model

TF-IDF + Logistic Regression

Initially, a traditional machine learning approach using TF-IDF Vectorization and Multinomial Logistic Regression was implemented.

TF-IDF converts text into numerical feature vectors based on word frequency and importance.

Although this approach is computationally efficient, it has limitations in understanding semantic meaning and contextual relationships.

Model Performance

Validation Accuracy: 0.2274

Test Accuracy: 0.2407

Classification Report

Class	Precision	Recall	F1-Score
pants-fire	0.14	0.28	0.19
false	0.28	0.22	0.24
barely-true	0.26	0.25	0.26
half-true	0.27	0.22	0.25
mostly-true	0.22	0.21	0.22
true	0.26	0.30	0.27

Overall Accuracy: 24%

Analysis

The TF-IDF model performed reasonably for some classes but struggled to capture deeper semantic relationships between similar statements.

Semantic Machine Learning Model

Sentence-BERT Embeddings + Softmax Logistic Regression

To improve contextual understanding, Sentence-BERT embeddings were used instead of traditional TF-IDF features.

Sentence-BERT converts sentences into dense semantic vectors that preserve contextual meaning.

The generated embeddings were then passed into a Softmax Logistic Regression classifier.

Model Performance

Validation Accuracy: 0.2336

Test Accuracy: 0.2336

Classification Report

Class	Precision	Recall	F1-Score
pants-fire	0.15	0.42	0.22
false	0.28	0.20	0.23
barely-true	0.25	0.22	0.23
half-true	0.26	0.20	0.22
mostly-true	0.23	0.22	0.22
true	0.25	0.27	0.26

Overall Accuracy: 23%

Analysis

Sentence-BERT *improved semantic* understanding compared to TF-IDF. However, the close similarity between LIAR labels made classification difficult.

Classes such as:

- half-true
- barely-true
- mostly-true

contain overlapping meanings, which affected model performance.

Deep Learning Model

DistilBERT Sequence Classification

The primary deep learning model used in this project is DistilBERT.

DistilBERT is a lightweight transformer model derived from BERT. It provides strong contextual understanding while requiring fewer computational resources.

The model was fine-tuned for six-class LIAR truthfulness classification.

Why DistilBERT?

DistilBERT was selected because:

- It understands contextual relationships better than traditional ML methods
- It performs efficiently on limited hardware
- It provides faster training compared to full BERT

- It is suitable for sequence classification task

Training Environment

The model was trained using Google Colab GPU support.

Model Performance

Test Accuracy: 0.2628

Classification Report

Class	Precision	Recall	F1-Score
pants-fire	0.24	0.05	0.09
false	0.28	0.38	0.32
barely-true	0.27	0.20	0.23
half-true	0.25	0.17	0.20
mostly-true	0.25	0.45	0.32
true	0.29	0.19	0.23

Overall Accuracy: 26%

Analysis

DistilBERT achieved the **highest performance** among all implemented models.

The model showed better contextual understanding and improved classification of complex statements.

However, performance remained challenging because:

- LIAR labels are highly overlapping
- Statements are short and ambiguous
- Some categories contain subtle differences

Despite moderate accuracy, the model demonstrated better semantic learning compared to traditional methods.

Risk Scoring System

Instead of depending only on the predicted label, the system uses soft probability distributions to calculate risk.

Each LIAR label is mapped to a credibility score and risk score.

Label	Credibility Score	Risk Score
true	100%	0%
mostly-true	80%	20%
half-true	60%	40%
barely-true	40%	60%
false	20%	80%
pants-fire	0%	100%

The final risk score is calculated using:

Final Risk = 90% Model Risk + 10% Linguistic Risk

This approach provides more stable and explainable predictions.

Linguistic Report Module

The project also includes a linguistic analysis module.

The module checks suspicious writing patterns such as:

- Excessive capital letters
- Emotional language
- Too many exclamation marks
- Clickbait phrases
- Urgent commands like “WATCH NOW”

The linguistic report is used as a supporting signal rather than the main prediction source.

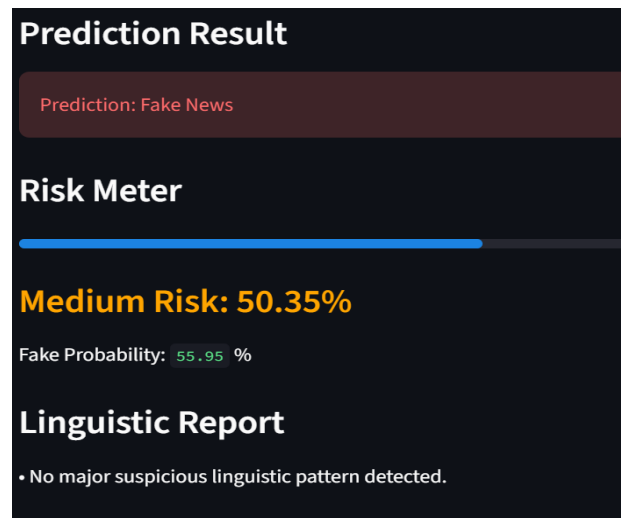
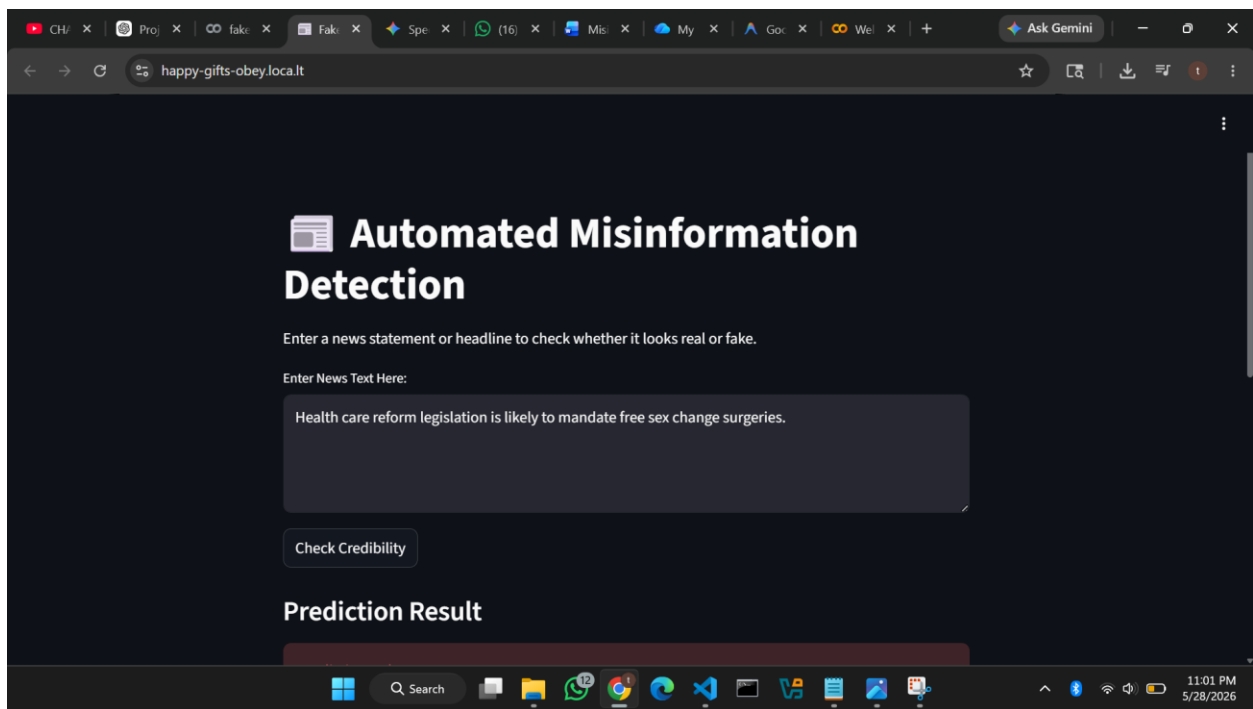
Streamlit User Interface

A Streamlit web application was developed for user interaction.

The interface allows users to:

- Enter a news statement or headline
- Generate credibility analysis
- View credibility score
- View risk score
- See final category output
- Read linguistic explanations

The interface was designed to be simple and user-friendly.



Project Workflow

The workflow of the system is as follows:

1. User enters a news headline or claim
2. Text preprocessing is performed
3. Model predicts LIAR truthfulness label
4. Prediction probabilities are generated
5. Soft risk score is calculated
6. Linguistic risk is added

7. Final credibility category is displayed
8. Streamlit interface shows results visually

Results and Evaluation

The results section is an important part of this project because it compares multiple NLP approaches for misinformation detection.

Three major models were implemented and evaluated:

- TF-IDF + Logistic Regression
- Sentence-BERT + Logistic Regression
- DistilBERT Deep Learning Model

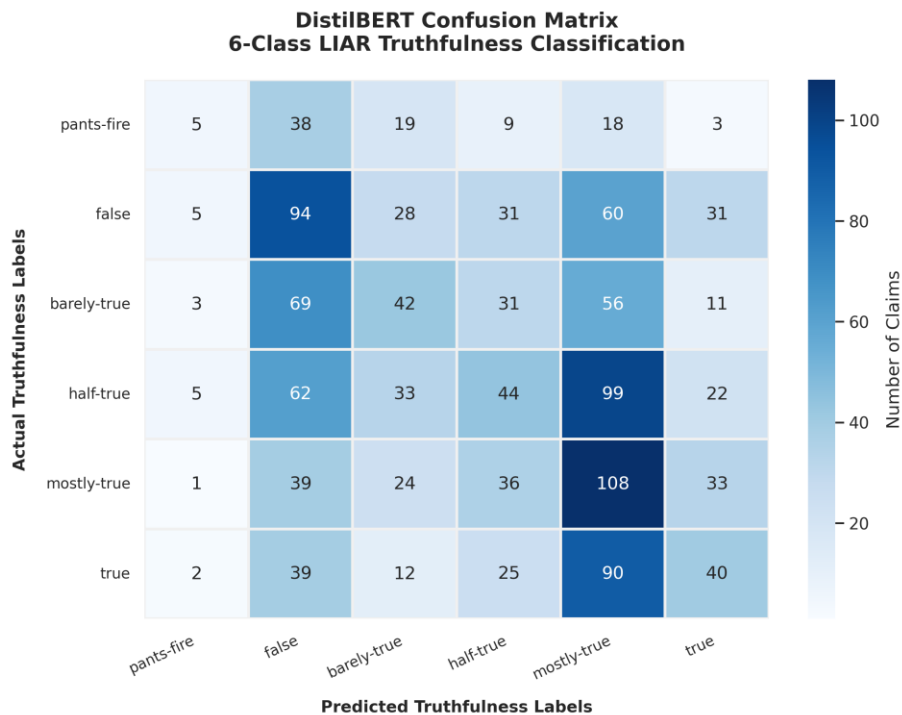
Comparative Performance

Model	Validation Accuracy	Test Accuracy
TF-IDF + Logistic Regression	22.74%	24.07%
Sentence-BERT + Logistic Regression	23.36%	23.36%
DistilBERT	—	26.28%

Observations

- DistilBERT achieved the highest accuracy.
- Sentence-BERT improved semantic understanding.
- Traditional TF-IDF struggled with contextual meaning.
- Overlapping LIAR labels reduced overall classification performance.
- Transformer-based models performed better than traditional approaches.

DistilBERT Confusion Matrix



Discussion

The confusion matrix shows that the model frequently confused neighboring truthfulness categories such as:

- mostly-true and half-true
- barely-true and false
- half-true and mostly-true

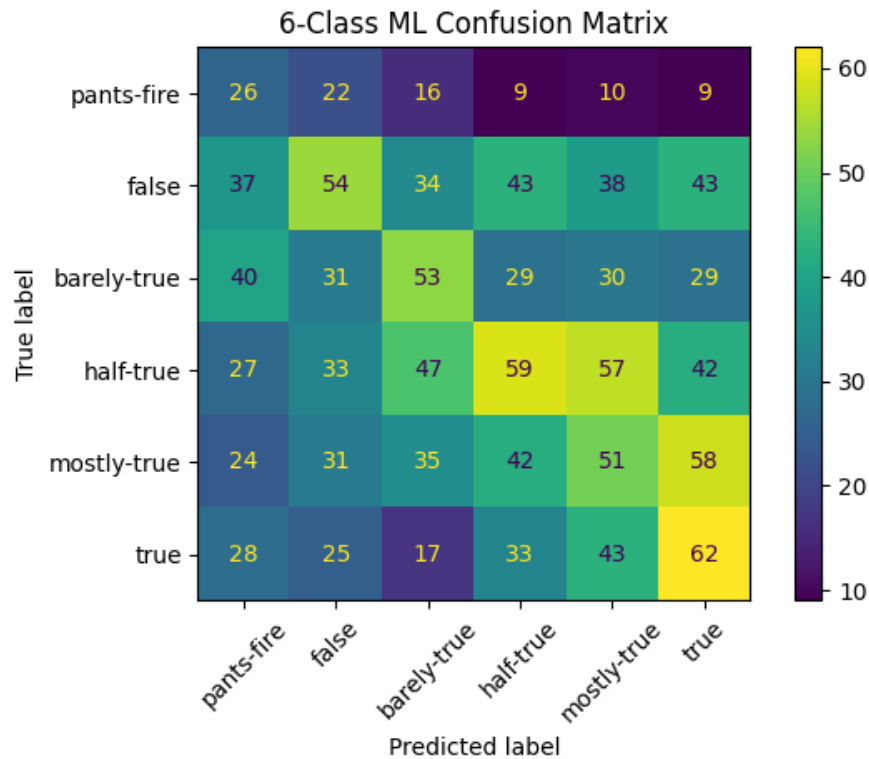
This behavior is expected because these labels contain semantic overlap.

The model performed comparatively better on:

- false
- mostly-true

while pants-fire remained difficult due to fewer examples and highly extreme language.

ML model TF-IDF + Logistic Regression Confusion Matrix



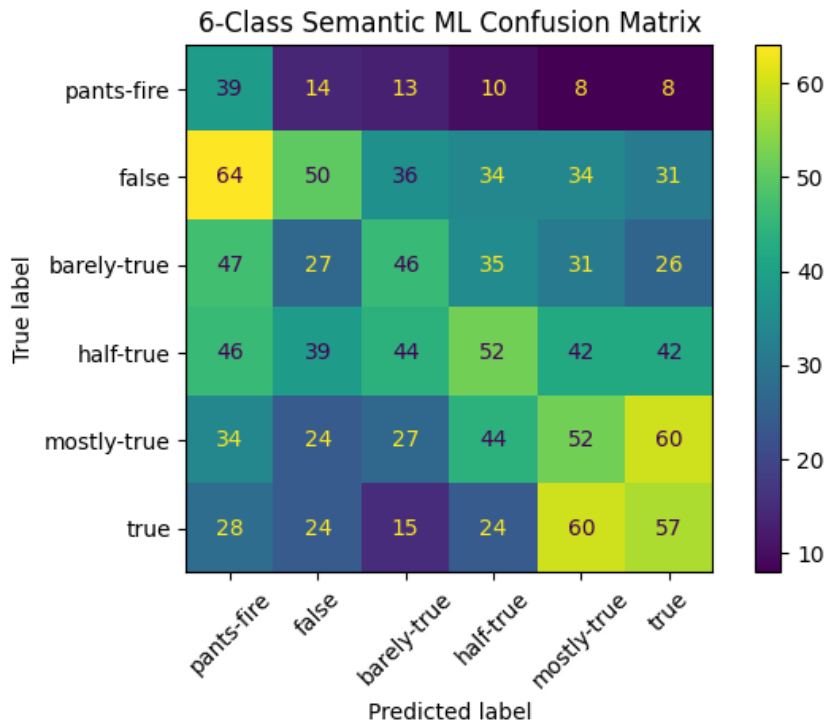
Analysis

The confusion matrix shows that the TF-IDF model struggled to clearly separate closely related truthfulness categories. Most misclassifications occurred between neighboring labels such as half-true, barely-true, and mostly-true.

The model relied mainly on word frequency features and could not fully understand contextual meaning. As a result, several statements with similar vocabulary but different meanings were classified incorrectly.

Although the model achieved moderate performance, it provided a useful baseline for comparing semantic embeddings and transformer-based approaches.

Semantic ML Confusion Matrix



Discussion

The semantic machine learning model showed improved contextual understanding compared to TF-IDF.

However, classification boundaries between neighboring labels remained challenging.

The model was still limited compared to transformer-based contextual learning.

Model Comparison Analysis

The project demonstrates that deep learning and transformer models are more effective for NLP-based misinformation detection compared to traditional ML methods.

Traditional approaches rely heavily on word frequency and statistical features.

Transformer models such as DistilBERT understand:

- semantic meaning
- context
- word relationships
- sentence structure

This makes them better suited for credibility analysis.

Although the achieved accuracy is moderate, the project successfully demonstrates:

- multi-class misinformation classification
- semantic understanding
- explainable risk scoring
- hybrid NLP architecture

Business Perspective

Market Need

The spread of fake news and misinformation has created demand for automated credibility analysis systems. Social media users, businesses, and fact-checking organizations require tools that can quickly identify suspicious information.

Target Users

Potential users include:

- Social media users
- Journalists
- Fact-checking organizations
- Educational institutions
- Businesses monitoring online reputation

Competitor Analysis

Existing platforms such as PolitiFact, Snopes, and FactCheck.org mainly depend on manual verification. This project focuses on automated NLP-based credibility prediction using transformer models and risk scoring.

Scalability

The system can be extended further using:

- larger transformer models
- multilingual support
- live fact-checking APIs

- source credibility analysis
- article-level verification

Future Enhancements

The project can be improved further by:

- Adding Retrieval-Augmented Generation (RAG)
- Integrating live fact-checking APIs
- Adding source credibility verification
- Supporting multilingual misinformation detection
- Using larger transformer architectures
- Adding explainable AI visualizations
- Improving training using larger datasets

Conclusion

This project presented a hybrid deep-learning approach for automated misinformation detection using Natural Language Processing techniques.

The system moved beyond simple binary fake news classification and implemented a six-class credibility framework using the LIAR dataset.

Multiple machine learning and deep learning approaches were explored, including TF-IDF, Sentence-BERT embeddings, and DistilBERT sequence classification.

Among all models, DistilBERT achieved the best performance due to its contextual understanding capabilities.

The project also introduced soft risk scoring and linguistic analysis to improve explainability and user understanding.

The final Streamlit application provides an interactive platform where users can analyze statements and receive credibility assessments.

Overall, the project demonstrates the practical use of NLP, semantic embeddings, and transformer models in addressing misinformation challenges.